

# CV Projects

Human Pose Estimation in Sports  
3D Pose Reconstruction

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# Projects for A.Y. 2025-2026



## What we expect:

- Look for state of art methods
- Implement and test
- Improve
- Read as much as you can
- Make assumptions
- Record videos
- Use pre-recorded data

# Projects for A.Y. 2025-2026



What we expect when ask for a meeting?

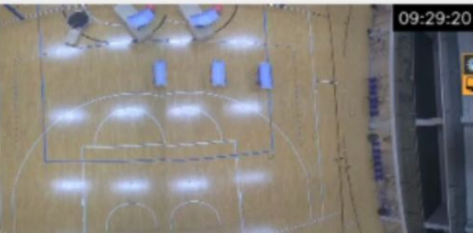
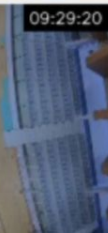
- Show that you tried
- Show difficulties encountered and actions taken to resolve

# Notes!

- Deep Learning is welcome, but not required (we cannot provide GPUs)
- Videos are for internal use only. Do not share them.
- The evaluation of your project depends more on your interaction with the tutor rather than on the numerical results
- The goal of the project is for you to learn something and face the real-world issues
- The project is individual for each group, but you are allowed to communicate, share ideas, share best practices. It is not a competition.
- Anyone does not have a group and would like to join one?

# Sanbapolis facility





# Project Goal and step

**Estimate the player's poses using the multiview camera setup at Sanbapolis, and evaluate the accuracy of your results.**

1. Annotate player's poses
2. Get the 3D position of the player from the triangulation of the cameras
  - a. Evaluate the accuracy of your triangulation by reprojecting the estimated 3D skeleton back into camera view and comparing the result with the ground truth 2D annotations
3. Improve the camera calibration by performing Bundle Adjustment on the points of the court
- a) Evaluate the accuracy of the triangulation with the improved camera parameters
4. Display the 3D Skeleton on Unreal

Team of 1 pp: task 1,2,2a,3 is mandatory. Bonus: task 3a.

Team of 2 pp: task 1,2,2a,3,3a is mandatory. Bonus: task 4.

Team of 3 pp: task 1,2,2a,3,3a,4 is mandatory. Bonus: task 4 for both triangulation at 2 and at 3

***The bonus task is NOT mandatory and is left to each student's discretion, as an additional application-oriented challenge.***

# 1) Annotate Skeletons

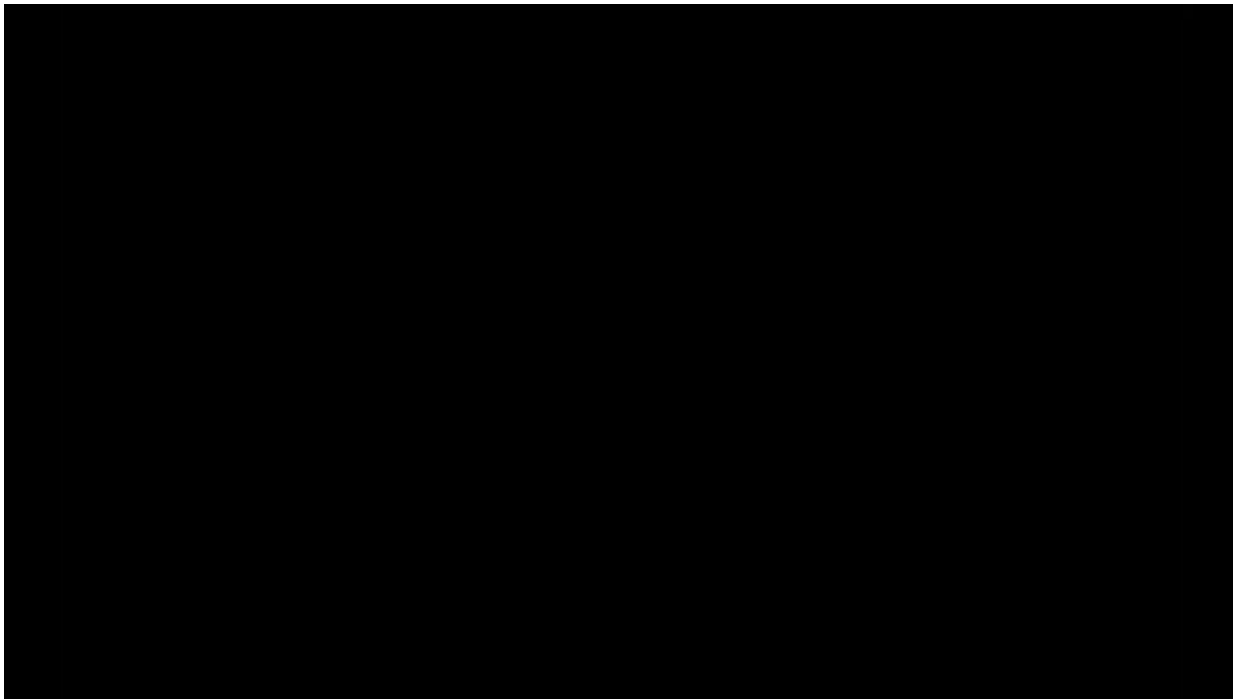
- There are 6 camera viewpoint for each action
  - One action per group
  - Around 30 frames per person
  - For each frames annotate one skeleton per player (both teams have to be annotated).
- 
- register to Roboflow (one account per group)
  - **send to the tutor one email per group to be added to Roboflow**
  - for testing and evaluation it is mandatory to use at least the action you have annotated. You can test and evaluate on actions annotated by other groups.



| COLOR ⓘ                                                                         | CLASS NAME | COUNT ↻ | KEYPOINTS                         |
|---------------------------------------------------------------------------------|------------|---------|-----------------------------------|
|  | Red_11     | 1       | 18 <a href="#">Edit Keypoints</a> |
|  | Red_13     | 1       | 18 <a href="#">Edit Keypoints</a> |
|  | Red_23     | 1       | 18 <a href="#">Edit Keypoints</a> |
|  | Red_7      | 1       | 18 <a href="#">Edit Keypoints</a> |
|  | Red_9      | 1       | 18 <a href="#">Edit Keypoints</a> |
|  | White_10   | 1       | 18 <a href="#">Edit Keypoints</a> |
|  | White_11   | 1       | 18 <a href="#">Edit Keypoints</a> |
|  | White_13   | 1       | 18 <a href="#">Edit Keypoints</a> |
|  | White_22   | 1       | 18 <a href="#">Edit Keypoints</a> |
|  | White_27   | 1       | 18 <a href="#">Edit Keypoints</a> |

- Draw a bounding box around the player
- Move the joints in correct position
- Assign to the skeleton the correct class (color\_number player shirt)
- Do it for all the players

Annotate player's poses



## 2) 3D player's position

Use the 2D annotations from multiview cameras and apply triangulation to get the 3D positions of the player

1. Rectified the video at input (code is provided. Remember: if you rectify the video you also have to apply the same transformation to the GT/the coordinates you have tracked).
2. Triangulation
3. Display 3D player's pose
4. Reproject the 3D skeleton back to camera view. Estimate the triangulation accuracy wrt ground truth annotations using common metrics (mean per joint position error, mean square error, ...)

### 3) Bundle Adjustment

- Bundle Adjustment refines the camera calibration by minimizing the reprojection error of known 3D points. Use the annotated court points and their known 3D positions to optimize the camera parameters. Then, repeat the triangulation of the player's joints with the updated calibration. Compare the new 3D reconstruction with the original one and analyze whether the improved calibration reduces the error. Reference paper (<https://eth-ait.github.io/WorldPoseDataset/>)

#### 3a)

- Estimate the triangulation accuracy with the new camera parameters wrt ground truth annotations using common metrics (mean per joint position error, mean square error, ...)

## 4) Unreal Engine

Display 3D data on Unreal. (contact tutor for more details)

# How do I deliver my project?



- Code + Readmen (push on Github with Github Classroom)
- Dataset
- Report with results (max 5 pages) (LNCS format, PDF)
- Link to the presentation in Google Slides format only
- Oral presentation on the day of the exam (8 min per group + Q&A)

**Deadline:** All material must be delivered to the lecturers **at least 3 working days before** the exam. No exceptions will be made.

# How do I deliver my project?



Accept the GitHub Classroom invitation

<https://classroom.github.com/a/sDy4Ysr0>

- This helps confirm the final groups
- It also lets me know if anyone wants to change topic
- Then I will create the dataset on Roboflow
- Each group will receive the images to annotate
- The material will be available from **Thursday, 23 April**

**NameSurname1/NameSurname2**

Computer-Vision-UniTN-Classroom

## Accept the group assignment — [CV25/26] - Player Tracking

Before you can accept this assignment, you must create or join a team. Be sure to select the correct team as you won't be able to change this later.

Join an existing team:

Antonio 0 students

Join

ConciMartinelli 0 students

Join

Or... Create a new team:

Super awesome team name

+ Create team

# Project's Score



Max project score: 15 points

- Code(5pt)
- Report (2pt)
- Supplementary (e.g sample video results) (1pt)
- Independence in carrying out a project (1pt)
- Oral (6pt)